MCQs on Number Systems

1. In a fixed-point number system, the position of the binary point is:

A. Changed during every arithmetic operation
B. Fixed at a predetermined position
C. Always placed after the least significant bit
D. Stored separately in memory

Answer: B. Fixed at a predetermined position

Explanation: In fixed-point representation, the binary point has an **implied fixed position** agreed upon when interpreting the bits.

2. An 8-bit fixed-point number has 4 integer bits and 4 fractional bits. Where is the implied binary point in 01101100?

A. 0.1101100
B. 01.101100
C. 011.01100
D. 0110.1100

Answer: D. 0110.1100

Explanation: Four bits are assigned to the integer part and four to the fractional part: 0110.1100

3. What is the decimal value of the unsigned fixed-point number 0110.1100?

A. 6.25
B. 6.50
C. 6.75
D. 6.875

Answer: C. 6.75

Explanation:

$$0110.1100$$
$$= 2^2 + 2^1 + 2^{-1} + 2^{-2}$$
$$= 4 + 2 + 0.5 + 0.25 = 6.75$$

4. In the fixed-point number 0010.0110, the weight of the rightmost bit is:

A. 2^{-1}
B. 2^{-2}
C. 2^{-3}
D. 2^{-4}

Answer: D. 2^{-4}

Explanation: With four fractional bits, their weights are:

$$2^{-1}, 2^{-2}, 2^{-3}, 2^{-4}$$

5. What is the decimal value of the fractional binary number 0.1000_2?

A. 0.25
B. 0.5

C. 0.75
D. 1.0

Answer: B. 0.5

Explanation:

$$0.1000_2 = 1 \times 2^{-1} = 0.5$$

6. What is the binary fixed-point representation of 0.75_{10} using four fractional bits?

A. 0000.0011
B. 0000.0110
C. 0000.1100
D. 0000.1110

Answer: C. 0000.1100

Explanation:

$$0.75 = 0.5 + 0.25$$
$$= 2^{-1} + 2^{-2}$$

Therefore:

$$0.75_{10} = 0000.1100_2$$

7. Which notation can be used to represent signed fixed-point numbers?

A. Only unsigned binary
B. Only BCD
C. Two's complement or sign-magnitude
D. Gray code only

Answer: C. Two's complement or sign-magnitude

8. In sign-magnitude representation, the most significant bit generally represents:

A. Fraction
B. Sign
C. Binary point
D. Least significant value

Answer: B. Sign

Explanation: Usually:

- 0 → positive
- 1 → negative

9. Which binary fraction represents 0.625_{10}?

A. 0.1000
B. 0.1010

C. 0.1100
D. 0.1110

Answer: B. 0.1010

Explanation:

$$0.1010_2 = 2^{-1} + 2^{-3}$$
$$= 0.5 + 0.125 = 0.625$$

10. In an 8-bit fixed-point representation with four fractional bits, the smallest fractional place value is:

A. 2^{-1}
B. 2^{-2}
C. 2^{-4}
D. 2^{-8}

Answer: C. 2^{-4}

$$2^{-4} = \frac{1}{16} = 0.0625$$

11. What is the decimal value of the unsigned fixed-point number 0101.1010_2?

A. 5.25
B. 5.5
C. 5.625
D. 5.75

Answer: C. 5.625

Explanation:

Integer part:

$$0101_2 = 5$$

Fraction:

$$.1010_2 = 0.5 + 0.125 = 0.625$$

Therefore:

$$5 + 0.625 = 5.625$$

12. What is the decimal value of 0011.0110_2 when interpreted as an unsigned fixed-point number?

A. 3.25
B. 3.375
C. 3.5
D. 3.75

Answer: B. 3.375

Explanation:

$$0011.0110$$

$$= 2 + 1 + 0.25 + 0.125$$
$$= 3.375$$

13. The absolute value $+2.375$is represented as:

0010.0110

What is the **sign-magnitude representation** of -2.375?

A. 0010.0110
B. 1010.0110
C. 1101.1010
D. 1110.0110

Answer: B. 1010.0110

Explanation: In sign-magnitude representation, only the sign bit changes:

$$0010.0110 \rightarrow 1010.0110$$

14. The positive fixed-point value $+2.375$ is:

0010.0110

What is its 8-bit two's complement representation for -2.375?

A. 1010.0110
B. 1101.1001
C. 1101.1010
D. 1110.0110

Answer: C. 1101.1010

Explanation:

Start with:

0010 0110

Invert:

1101 1001

Add 1:

1101 1010

Therefore:

$$-2.375 = 1101.1010$$

15. Which statement correctly distinguishes sign-magnitude from two's complement?

A. Both are always identical for negative numbers
B. Sign-magnitude changes the sign bit, while two's complement involves inversion and adding 1
C. Two's complement changes only the sign bit
D. Sign-magnitude requires inversion and adding 1

Answer: B

16. What is the fixed-point binary representation of 0.375_{10} using four fractional bits?

A. 0000.0011
B. 0000.0110
C. 0000.1010
D. 0000.1100

Answer: B. 0000.0110

Explanation:

$$0.375 = 0.25 + 0.125$$
$$= 2^{-2} + 2^{-3}$$

Therefore: $0.375 = 0.0110_2$

17. Consider the fixed-point number:

0010.1011

What is its unsigned decimal value?

A. 2.625
B. 2.6875
C. 2.75
D. 2.8125

Answer: B. 2.6875

Explanation:

$$0010.1011$$
$$= 2 + 0.5 + 0.125 + 0.0625$$
$$= 2.6875$$

18. Why is two's complement generally preferred for signed fixed-point arithmetic?

A. It eliminates fractional bits
B. It allows signed addition to be performed using ordinary binary addition
C. It requires no sign bit
D. It increases the number of bits automatically

Answer: B

Explanation: Two's complement allows positive and negative numbers to be added using the **same binary addition circuitry**.

19. In an 8-bit fixed-point format with four fractional bits, what bit pattern represents $+0.75$?

A. 0000 0011
B. 0000 0110
C. 0000 1100
D. 0001 1000

Answer: C. 0000 1100

Because:

$$0.75 = 0.1100_2$$

20. In an 8-bit fixed-point system with 4 integer bits and 4 fractional bits, $+0.625$is represented as:

0000.1010

What is -0.625in two's complement?

A. 1000.1010
B. 1111.0101
C. 1111.0110
D. 1000.0110

Answer: C. 1111.0110

Explanation:

Positive:

0000 1010

Invert:

1111 0101

Add 1:

1111 0110

Therefore:

$$-0.625 = 1111.0110$$

21. Perform the fixed-point addition:

$$0000.1100 + 1111.0110$$

using 8-bit two's complement arithmetic.

A. 0000.0010
B. 0000.0110
C. 1111.0010
D. 10000.0010

Answer: A. 0000.0010

Explanation:

The numbers represent:

$$0000.1100 = +0.75$$
$$1111.0110 = -0.625$$

So:

$$0.75 - 0.625 = 0.125$$

and:

$$0.125 = 0000.0010$$

Binary addition produces a carry beyond the 8-bit result, which is **discarded**.

22. A bit pattern 01101100 is stored in memory. Can we determine its fixed-point decimal value without knowing the binary-point position?

A. Yes, it is always 108
B. Yes, it is always 6.75
C. No, the binary-point position must be known
D. No, fixed-point values cannot be converted to decimal

Answer: C

Explanation: The bits themselves do not store the binary point.

For example:

01101100

could be interpreted as:

0110.1100

or

011.01100

or

01.101100

Each interpretation gives a different numerical value.

23. Suppose the bit pattern is:

00110100

If interpreted with 4 fractional bits, it becomes:

0011.0100

What is its decimal value?

A. 3.125
B. 3.25
C. 3.5
D. 3.75

Answer: B. 3.25

Explanation:

$$0011.0100$$
$$= 3 + 2^{-2}$$
$$= 3 + 0.25 = 3.25$$

24. The same bit pattern 00110100 is now interpreted with only 2 fractional bits. What is its value?

A. 3.25
B. 6.5
C. 13
D. 52

Answer: C. 13

Explanation:

With two fractional bits:

001101.00

$$001101_2 = 13$$

Therefore:

$$001101.00_2 = 13$$

This demonstrates that the **same bits can represent different values depending on the assumed binary-point position**.

25. Which of the following correctly represents -2.375in 8-bit fixed-point two's complement, assuming four fractional bits?

A. 1010.0110
B. 1101.1010
C. 0010.0110
D. 1101.0110

Answer: B. 1101.1010

26. An 8-bit fixed-point two's complement number has four fractional bits. What is the value of the least significant bit?

A. 1
B. 0.5
C. 0.125
D. 0.0625

Answer: D. 0.0625

Explanation:

The rightmost bit has weight:

$$2^{-4} = \frac{1}{16} = 0.0625$$

This is also the **resolution/step size** of this fixed-point format

27. Which statement about fixed-point numbers is FALSE?

A. The binary point is usually implied rather than physically stored
B. Two's complement can represent negative fixed-point numbers
C. The interpretation of the bits depends on the agreed binary-point position
D. The location of the binary point can always be determined by examining only the stored bits

Answer: D

Explanation: A fixed-point number is physically just a **collection of bits**. The bit pattern itself does not tell us where the implied binary point is located.

28. The main advantage of floating-point representation over fixed-point representation is that floating-point can:

A. Represent only integers
B. Represent a much wider range of very large and very small numbers
C. Eliminate the need for binary numbers
D. Represent only positive numbers

Answer: B

Explanation

In fixed-point representation, the number of integer and fractional bits is fixed.

Floating-point allows the binary point to effectively **"float"** by using an exponent.

Therefore, it can represent both:

- Very large numbers
- Very small numbers

29. A floating-point number is analogous to:

A. Boolean algebra
B. Scientific notation
C. Gray code
D. Two's complement only

Answer: B

Explanation

Consider decimal scientific notation:

$$4500 = 4.5 \times 10^3$$

Similarly, binary floating-point notation can be written as:

$$10100_2 = 1.0100_2 \times 2^4$$

Thus, floating-point representation works similarly to **scientific notation**.

30. IEEE 754 single-precision floating-point representation uses how many bits?

A. 16 bits
B. 32 bits
C. 48 bits
D. 64 bits

Answer: B. 32 bits

1+8+23=32 bits

1 – sign bit

8 – exponent

23 – Fraction

31. What does a sign bit of 0 indicate for an ordinary IEEE 754 floating-point number?

A. Negative
B. Positive
C. Infinity
D. NaN

Answer: B. Positive

Explanation

The sign convention is:

$$0 \rightarrow \text{positive}$$
$$1 \rightarrow \text{negative}$$

32. What is the bias used in IEEE 754 single-precision floating-point representation?

A. 63
B. 127
C. 128
D. 1023

Answer: B. 127

Explanation

For normal single-precision numbers:

$$E_{\text{biased}} = E_{\text{actual}} + 127$$

33. If the actual exponent is -3, what is the biased exponent?

A. 124
B. 127
C. 130
D. 131

Answer: A. 124

Explanation

$$E_{\text{biased}} = -3 + 127$$
$$= 124$$

In binary:

$$124_{10} = 01111100_2$$

34. Convert 20_{10}into normalized binary scientific notation.

A. 1.01×2^4
B. 1.01×2^3
C. 10.1×2^4
D. 1.1×2^5

Answer: A

Explanation

First convert 20 to binary:

$$20_{10} = 10100_2$$

Normalize:

$$10100_2 = 1.0100_2 \times 2^4$$

Therefore:

$$\boxed{1.01_2 \times 2^4}$$

35. What is the normalized binary representation of 12_{10}?

A. 1.1×2^3
B. 1.1×2^2
C. 1.01×2^3
D. 1.11×2^2

Answer: A

Explanation

$$12_{10} = 1100_2$$

Move the binary point:

$$1100_2 = 1.100_2 \times 2^3$$

Therefore:

$$\boxed{1.1_2 \times 2^3}$$

36. The number $\mathbf{1.101_2 \times 2^4}$ has an actual exponent of:

A. 1
B. 4
C. 101
D. 131

Answer: B. 4

Explanation

In:

$$M \times 2^E$$

the exponent is E.

Therefore:

$$E = 4$$

Its **biased exponent** would be:

$$4 + 127 = 131$$

Do not confuse **actual exponent = 4** with **biased exponent = 131**.

37. Why is the leading 1 of a normalized binary floating-point number usually not stored?

A. It is always 0
B. It is always 1 for normalized nonzero binary numbers
C. It represents the sign
D. It represents the exponent

Answer: B

Explanation

A normalized binary number has the form:

$$1.xxxxx \times 2^E$$

The digit before the binary point is always:

$$1$$

Therefore, storing it would waste one bit.

It is assumed automatically and is called the:

implicit leading one

38. What is the biased exponent for an actual exponent of $+10$?

A. 117
B. 127
C. 137
D. 138

Answer: C. 137

Explanation

$$10 + 127 = 137$$

In binary:

$$137_{10} = 10001001_2$$

39. An IEEE 754 single-precision number has biased exponent 130. What is its actual exponent?

A. 3
B. −3
C. 130
D. 257

Answer: A. 3

Explanation

To recover the actual exponent:

$$E = E_{\text{biased}} - 127$$

Therefore:

$$130 - 127 = 3$$

40. An IEEE 754 single-precision number has biased exponent 10000010_2. What is the actual exponent?

A. 2
B. 3
C. 129
D. 130

Answer: B. 3

Explanation

First convert:

$$10000010_2 = 130_{10}$$

Then subtract the bias:

$$130 - 127 = 3$$

Therefore:

$$\boxed{E = 3}$$

41 What is the normalized binary representation of 40_{10}?

A. 1.01×2^5
B. 1.01×2^4
C. 1.1×2^5
D. 1.001×2^5

Answer: A

Explanation

Convert 40 to binary:

$$40_{10} = 101000_2$$

Normalize:

$$101000_2 = 1.01000_2 \times 2^5$$

Therefore:

$$\boxed{1.01_2 \times 2^5}$$

42. For $40_{10} = 1.01_2 \times 2^5$, what is the biased exponent?

A. 5
B. 127
C. 130
D. 132

Answer: D. 132

Explanation

Actual exponent:

$$E = 5$$

Bias:

$$127$$

Therefore:

$$5 + 127 = 132$$

And:

$$132_{10} = 10000100_2$$

43. Which is the correct IEEE 754 representation of $+\mathbf{40_{10}}$?

A. 0 10000100 01000000000000000000000
B. 0 00000101 01000000000000000000000
C. 1 10000100 01000000000000000000000
D. 0 10000100 10100000000000000000000

Answer: A

Explanation

Step 1:

$$40 = 101000_2$$

Step 2: Normalize:

$$1.01000 \times 2^5$$

Step 3: Sign:

$$0$$

Step 4: Biased exponent:

$$5 + 127 = 132 = 10000100$$

Step 5: Remove implicit leading 1:

$$1.\boxed{01000}$$

Fraction:

01000000000000000000000

Final:

$$\boxed{0\ 10000100\ 01000000000000000000000}$$

44. Convert 18.5_{10} to normalized binary notation.

A. 1.00101×2^4
B. 1.01001×2^4
C. 1.00101×2^5
D. 1.101×2^3

Answer: A

Explanation

Integer part:

$$18_{10} = 10010_2$$

Fraction:

$$0.5 = 0.1_2$$

Therefore:

$$18.5 = 10010.1_2$$

Normalize:

$$10010.1 = 1.00101 \times 2^4$$

45 Which is the correct IEEE 754 single-precision representation of -18.5?

A. 1 10000011 00101000000000000000000
B. 0 10000011 00101000000000000000000
C. 1 00000100 00101000000000000000000
D. 1 10000011 10010100000000000000000

Answer: A

Explanation

We found:

$$18.5 = 10010.1_2$$

Normalize:

$$1.00101 \times 2^4$$

Since negative:

$$S = 1$$

Biased exponent:

$$4 + 127 = 131$$
$$131 = 10000011_2$$

Fraction after removing the implicit 1:

$$00101$$

Pad to 23 bits:

00101000000000000000000

Therefore:

$$\boxed{1\ 10000011\ 00101000000000000000000}$$

46. Decode:

1 10000001 01000000000000000000000

A. −2.5
B. −5
C. +5
D. −10

Answer: B. −5

Explanation

Sign:

$$S = 1 \Rightarrow \text{negative}$$

Exponent:

$$10000001_2 = 129$$

Actual exponent:

$$129 - 127 = 2$$

Fraction:

010000...

Restore implicit leading 1:

$$1.01_2$$
$$1.01_2 = 1.25$$

Therefore:

$$1.25 \times 2^2 = 5$$

Apply sign:

$$\boxed{-5}$$

47. A normalized binary number is:

$$1.01101_2 \times 2^{-4}$$

What is its biased exponent in single precision?

A. 123
B. 124
C. 127
D. 131

Answer: A. 123

Explanation

$$E = -4$$

Therefore:

$$E_{\text{biased}} = -4 + 127$$
$$= 123$$

In binary:

$$123 = 01111011_2$$

48. Convert 10.25_{10} to normalized binary notation.

A. 1.01001×2^3
B. 1.00101×2^3
C. 1.0101×2^3
D. 1.01001×2^4

Answer: A

Explanation

Integer:

$$10 = 1010_2$$

Fraction:

$$0.25 = 0.01_2$$

Therefore:

$$10.25 = 1010.01_2$$

Normalize:

$$1010.01 = 1.01001 \times 2^3$$

49 In IEEE 754 single precision, which exponent pattern is reserved for zero and subnormal numbers?

A. All 0s
B. All 1s
C. 10000000
D. 01111111

Answer: A. All 0s

Explanation

An exponent field of:

00000000

is reserved for:

- Zero
- Subnormal numbers

For exact zero, the fraction is also all zeros.

50. Which exponent pattern is used for infinity and NaN?

A. All 0s
B. All 1s
C. Alternating 1s and 0s
D. 01111111

Answer: B. All 1s

Explanation

Exponent:

11111111

is reserved for special values.

If fraction = all 0s:

$$\pm\infty$$

If fraction ≠ 0:

$$NaN$$

51. Approximately how many significant decimal digits does double precision provide?

A. 7
B. 8
C. 15
D. 64

Answer: C. About 15 digits

52 A student sees the exponent field 10000101_2 in a normal single-precision number and concludes that the actual exponent is 133. What is the correct actual exponent?

A. 6
B. 127
C. 133
D. 260

Answer: A. 6

Explanation

First:

$$10000101_2 = 133$$

But 133 is the **biased exponent**, not the actual exponent.

Therefore:

$$E = 133 - 127$$
$$\boxed{E = 6}$$

This is an important distinction:

$$\boxed{\text{Stored exponent} \neq \text{Actual exponent}}$$

for normal IEEE 754 numbers.

53 The IEEE 754 representation of a normal number begins:

0 10000100 10100000000000000000000

What is its decimal value?

A. 26
B. 52
C. 13
D. 104

Answer: B. 52

Explanation

Exponent:

$$10000100_2 = 132$$

Actual exponent:

$$132 - 127 = 5$$

Fraction:

101...

Restore implicit 1:

$$1.101_2$$

Convert:

$$1.101_2 = 1 + 0.5 + 0.125 = 1.625$$

Now:

$$1.625 \times 2^5$$
$$= 1.625 \times 32$$
$$\boxed{52}$$